

■ Affine Cipher

- Extension of the shift cipher: rather than just adding the key to the plaintext, we also multiply by the key
- We use for this a key consisting of two parts: $k = (a, b)$

Let $k, x, y \in \{0, 1, \dots, 25\}$

- Encryption: $y = e_k(x) \equiv a x + b \pmod{26}$
- Decryption: $x = d_k(y) \equiv a^{-1}(y - b) \pmod{26}$

- Since the inverse of a is needed for inversion, we can only use values for a for which:

$$\gcd(a, 26)$$

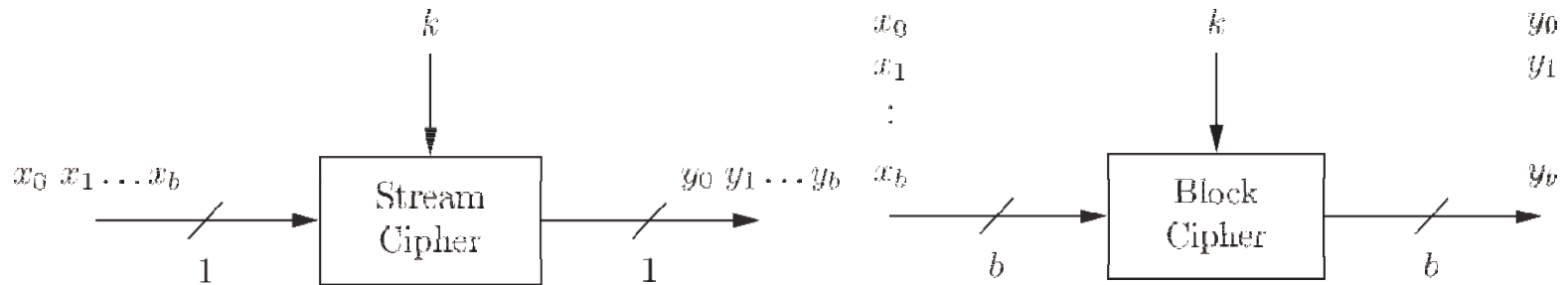
=1 There are 12 values for a that fulfill this condition.

- From this follows that the key space is only $12 \times 26 = 312$

Again, several attacks are possible, including:

- Exhaustive key search and letter frequency analysis, similar to the attack against the substitution cipher

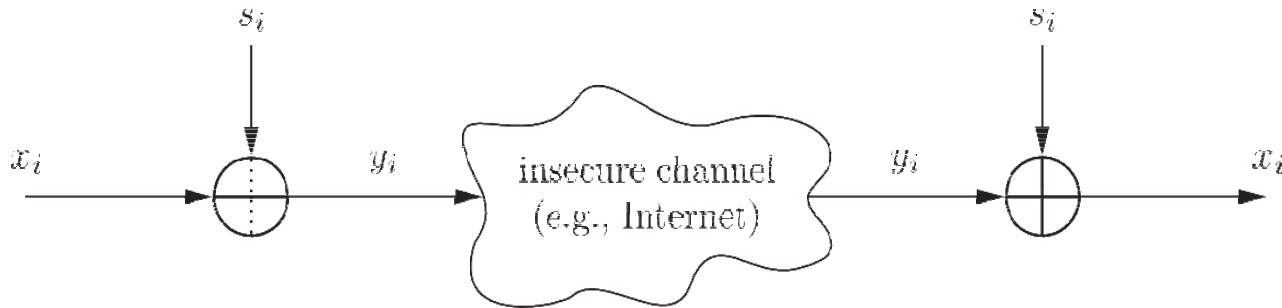
■ Stream Cipher vs. Block Cipher



- **Stream Ciphers**
 - Encrypt bits individually
 - Usually small and fast → common in embedded devices (e.g., A5/1 for GSM phones)
- **Block Ciphers:**
 - Always encrypt a full block (several bits)
 - Are common for Internet applications

■ Encryption and Decryption with Stream Ciphers

Plaintext x_i , ciphertext y_i and key stream s_i consist of individual bits



- Encryption and decryption are simple additions modulo 2 (aka XOR)
- Encryption and decryption are the same functions
- **Encryption:** $y_i = e_{s_i}(x_i) = x_i + s_i \bmod 2$ $x_i, y_i, s_i \in \{0,1\}$
- **Decryption:** $x_i = e_{s_i}(y_i) = y_i + s_i \bmod 2$